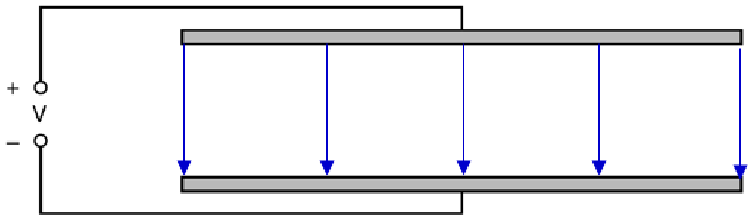
interference pattern. **B1**

5 **ai** An electric field is a region of space where a stationary charge experiences a force. **B1**

aii



Straight arrows pointing vertical downwards **B1**

Equally spaced field lines (at least 4 lines) **B1**

**If draw curve edges, need at least 4 lines excluding curve edges.*

aiii 1. $V = \dfrac{W}{Q} \quad \Rightarrow \quad W = QV$ **B1**

The electric field is acting vertical downwards. The electric force on the positive charge is also downwards. In moving the positive charge against the field, positive work is done by an external agent.

No mark awarded if there is a negative sign attached to *QV*.

aiii 2. Work done $W = Fd$ **B1**

No mark awarded if there is a negative sign attached to *Fd*.

aiv From part **(iii)**, we have $Fd = QV$

$$\frac{F}{Q} = \frac{V}{d}$$ **M1**

Since electric field strength is defined as the *force per unit positive charge* $\left(\frac{F}{Q}\right)$, **M1**

it is also numerically equals to the potential gradient $\left(\frac{V}{d}\right)$ **A0**

b The neutral point lies between the two positive charges and is closer to +1.0 μC.

Let x = distance of neutral point from the +1.0 μC charge and a positive value.

The net electric field strength is zero at the neutral point.

$E_1 - E_2 = 0$

$E_1 = E_2$

$$\frac{1}{4\pi\varepsilon_o}\frac{Q_1}{x^2} = \frac{1}{4\pi\varepsilon_o}\frac{Q_2}{(d-x)^2}$$ **C1**

$$\frac{1.0}{x^2} = \frac{4.0}{(6-x)^2} \rightarrow +\sqrt{\frac{1.0}{x^2}} = +\sqrt{\frac{4.0}{(6-x)^2}}$$ **C1**

$x = 2.0$ cm **A1**

6 **ai** $\frac{hc}{} \quad (6.63\times10^{-34})(3.00\times10^{8})$